Stealthy shorts: Informed liquidity supply

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DOES INEQUALITY AFFECT THE NEEDS OF THE POOR?

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Abstract
We investigate how inequality affects what is considered most necessary to purchase. Combining detailed consumption surveys in India with an instrumental variable approach, we first provide evidence that inequality is associated with a decrease in calorie consumption for poor households. This effect is driven by upward comparisons and associated with poor households spending more on luxuries, consistent with the relative deprivation theory. We then estimate a structural demand system to isolate the impact of inequality on needs from supply-side effects. We find that inequality increases the need of the poor for "little luxuries" (e.g., dairy products, energy use) at the expense of necessities (e.g., cereals). This shift in the hierarchy of needs accounts for more than two-thirds of the decline in the calorie consumption of the poor over the period. (JEL: D01, D12, I14, I30, O12)

1. Introduction

Since Engel's law was proposed in 1857, economists have developed empirical models of demand to classify goods as either necessities or luxuries.[1] For instance, adequate nutrition is widely considered as one of the most basic necessities.[2] Little is known,

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1. This classification relies on the estimation of income elasticities, which is a recurrent exercise in the literature (e.g., Chai and Moneta 2010; Almås 2012).

2. Engel's law states that "the poorer the individual, the greater must be the percentage of the income necessary for the maintenance of physical sustenance" (Engel 1857).

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however, about *what* makes a good a necessity or a luxury (Heffetz 2011, 2018). Inequality has been suspected to shift the needs of the poor toward the consumption patterns of richer individuals (Hopkins 2024), as the relatively deprived unfavorably compare their circumstances with richer others (Runciman 1966).[3] If this proves to be true, inequality may change the hierarchy of needs for the poor at the expense of nutritional ones.

Relative poverty posits that consumption standards are relative, so that inequality affects the cost-of-living of the relatively deprived (Ravallion and Chen 2011). However, it remains unclear whether this effect is due to a shift in needs, or to changes in consumers' budget constraint and supply environment. We call "needs" the necessary quantities that must be purchased regardless of prices and own income. Rather than affecting needs, inequality may render luxury products relatively cheaper or more easily available (Jaravel 2019), or affect the permanent income of the poor. Moreover, even if supply-side explanations were to explain only a fraction of consumer expenditure allocation (Dubois, Griffith, and Nevo 2014; Allcott et al. 2019), inequality can still be endogenous to consumer preferences.

This paper makes three primary contributions to the literature. First, we isolate the causal effect of inequality on basic needs. We tackle the endogeneity issue between consumption choices and inequality, and we structurally identify need-based parameters separately from income and local prices. Second, we provide a testable prediction of the effect of inequality on the hierarchy of needs. We call "relative necessities" the goods that become more necessary (income inelastic) to the relatively deprived. Third, we empirically find that relative necessities are also absolute luxuries.[4] Inequality shifts the hierarchy of needs toward goods more consumed by the affluent. We quantify a substantial impact of this shift on the nutritional intake of the poor in India.

Our setting is the consumption of the poor in India where, given widespread malnutrition, inequality may shift the hierarchy of needs at a significant nutritional cost. Three main features make this setting particularly suitable for our analysis. Firstly, the Indian National Sample Surveys (NSS) on consumption used in this paper include a sizable fraction of households below the official Indian poverty line, and provide a unique and highly detailed record of their expenditure and quantities consumed. Secondly, over the span of our study, India experienced major shocks in its income distribution, particularly following the 1991 economic liberalization reforms (Aghion et al. 2008; Topalova 2010; Chancel and Piketty 2019). Lastly, India exemplifies the potential existence of competing needs beyond adequate nutrition. The literature provides evidence in the Indian context that poor and malnourished individuals may divert a significant fraction of their income to satisfy other goals such as taste, culture,

3. Runciman (1966) defines a person as relatively deprived of X when (i) they do not have X, (ii) they see some other person or persons, which may include themselves at some previous or expected time, as having X, (iii) they want X, and (iv) they see it as feasible that they should have X.

4. Luxuries in our paper are day-to-day expenditure categories whose income elasticity is above 1.

or entertainment (Deaton and Subramanian 1996; Banerjee and Duflo 2007; Atkin 2013, 2016).

We build on the relative deprivation literature and define inequality exposure of a household as the normalized sum of the gaps in total expenditure between them and all other richer households in their district (Yitzhaki 1979; Clark and D'Ambrosio 2015). We first provide evidence of the overall effect of inequality exposure on consumption using the simple measure of calorie intake in a traditional Engel curves framework. Non-parametric Engel curves show a calorie loss for poor households more exposed to inequality at all levels of total expenditure. They also show a shift in expenditure from caloric goods and necessities to non-caloric goods and luxuries with inequality.

The endogeneity of total expenditure and inequality exposure is the primary identification challenge. First, consumption choices may be correlated with omitted household or district-level factors that also affect inequality exposure. Second, there is the possibility of reverse causation between consumption choices and inequality, for instance, if luxurious tastes at the bottom of the distribution lead to malnutrition, lowering the working capacity of the poor. Third, the typical measurement error issue affects category expenditure, total expenditure, and inequality exposure. To tackle these issues, we develop instrumental variables (IV) for total expenditure and inequality that (i) do not emerge from households' deliberate choices, and (ii) are not derived from the present characteristics of households' districts.

We rely on two key features of the Indian context. First, occupational choice is still largely determined by traditional occupations rigidly tied to the caste and religious affiliations given at birth (Cassan, Keniston, and Kleineberg 2021; Oh 2023). Second, the period that followed the liberalization reforms was associated with sudden and drastic changes in nationwide occupational returns unrelated to local labor market conditions.[5] We instrument total expenditure and inequality by shift-share instruments (Goldsmith-Pinkham, Sorkin, and Swift 2020; Borusyak, Hull, and Jaravel 2021) based on national returns to the historical (1987–1988) district occupational mix of the household's own caste/religious group (Atkin, Colson-Sihra, and Shayo 2021). In essence, individuals are expected to attain a higher income (vs. be exposed to higher levels of inequality) if they (vs. other households) are born into a caste or religion with a historical local prominence in occupations that reaped substantial returns post-liberalization at the national level.

We estimate Engel curves parametrically taking into account confounding factors such as demographics and occupations, and instrumenting for total expenditure and inequality exposure. The instrumented calorie Engel curve estimation leads to similar estimates than the ordinary least squares (OLS) estimation. The results are also consistent with the relative deprivation theory. First, using placebo measures of inequality exposure, we test which part of the income distribution drives the effect. We show that the calorie loss is driven by upward (not downward) comparisons with

5. The major impacts that the 1991 liberalization reforms had on the distribution of national income across occupations is well-documented (Aghion et al. 2008; Topalova 2010; Chancel and Piketty 2019).

richer others. Second, we confirm that inequality exposure increases the consumption of goods commonly consumed by the rich (luxuries).

We then present a structural demand system in which the need of each good has a basic component and a relative component that varies with inequality exposure. We structurally estimate our model on 12 disaggregated consumption categories that report quantities purchased. We use unit values as price proxies. These 12 categories capture above 80% of the total budget of the poor households in our data. We instrument for inequality exposure and total expenditure, as well as for prices by using prices in a nearby village in another district (Atkin 2013). We check that the estimation captures a basic necessary level of consumption that is consistent with physiological needs and affordable, representing 30% of the average budget of the poor.

We confirm that relative necessities are typically luxuries, with an estimated basic need, which is low or insignificant. In the Indian context, relative necessities are goods associated with social inclusion and shared in social settings (dairy products, energy use, spices, processed food and drinks, and meat) and, to a lesser extent, temptation or hedonic goods (sweets, intoxicants). Conversely, necessities, which are also the most nutritious (cereals, vegetables), become less necessary with inequality. We observe a drastic shift in the resulting income elasticities such that in the most unequal districts, spices and processed food and drinks actually become normal or necessary goods. Our results are robust to the estimation over the entire population, and to a generalization of our demand system more flexible in the price space.

Lastly, to illustrate the magnitude of the need-based channel, we show that our estimates are able to explain a significant fraction of the caloric consumption puzzle, that is, the fact that caloric consumption declined over time in India even among the poor (Deaton and Drèze 2009). We find that an increase of 1 standard deviation (SD) in inequality exposure is associated with 186 fewer calories per capita per day. This amounts to nearly one and a half adult portions of cooked rice (100g, 130 calories). If they would consume these "lost" calories, the fraction of poor Indian households with malnutrition would be on average 30 percentage points lower. This substantial calorie loss accounts for more than two-thirds of the mean calorie decline over time for poor households. The stability of the calorie loss estimates across Engel curves and structural analysis underscores that the effect of inequality on consumption at the bottom of the distribution is primarily demand-driven.

In the final section of the paper, we explore three alternative explanations other than relative deprivation that may explain the estimated effect of inequality on basic needs: credit markets, gift giving behavior, and access to food security networks. We find that none of those three alternatives are likely to explain our results.

Our paper contributes to two distinct strands of the literature: consumer theory and inequality. First, incorporating inequality within the scope of consumer theory contributes to our understanding of consumer expenditure allocation. A significant share of the observed variance in consumer choices remains unexplained by the usual economic determinants—prices and income—and is driven instead by persistent differences in preferences (Bronnenberg, Dubé, and Gentzkow 2012; Atkin 2013; Dubois, Griffith, and Nevo 2014; Allcott et al. 2019; Handbury 2021). The demand literature has introduced insights from social theory, such as peer effects (De Giorgi, Frederiksen, and Pistaferri 2020; Lewbel et al. 2022), culture (Atkin 2016; Jayachandran and Pande 2017), and identity choice (Atkin, Colson-Sihra, and Shayo 2021) to account for this variance. Recently, Bramoullé and Ghiglino (2022) explore the theoretical implications of lying below the neighbors' average consumption on income elasticities. Closer to ours, Heffetz (2011, 2018), shows that visible goods have a higher income elasticity, as richer individuals spend relatively more on them to signal status. We show that such goods could also be relative necessities, as poorer individuals consider them more necessary when inequality increases.

Second, including inequality as a determinant of basic needs provides empirical support to the theoretical literature on inequality as relative deprivation (Runciman 1966; Ravallion and Chen 2011; Hopkins 2024).[6] Although the link between inequality and consumption has been theorized early (Duesenberry 1949; Runciman 1966; Frank, Levine, and Dijk 2014), empirical evidence have emerged only recently (Charles, Hurst, and Roussanov 2009; Khamis, Prakash, and Siddique 2012; Bertrand and Morse 2016; Bursztyn et al. 2017; Bellet 2024). However, the existing literature provide little evidence on whether and how inequality affects the very poor in society,[7] and in the absence of price, information is unable to isolate a demand channel. The lack of evidence on inequality and consumption at the bottom of the distribution is not merely a result of data constraints. Initial theoretical frameworks conceptualizing inequality through the lens of a signaling game, focusing on ordinal preferences, provided limited justification for its influence on the poor (Hopkins and Kornienko 2004; Hopkins 2024). Our context and demand estimation are unique in their ability to quantify the causal impact of inequality on basic needs.

The article is organized as follows: Section 2 provides the background to relative needs and relative deprivation theory. Section 3 describes the data and variables. Section 4 presents evidence of the link between inequality and consumption of the poor in a traditional Engel curves framework. Section 5 introduces our model of relative needs. Section 6 structurally estimates the need-based channel of inequality, identifies relative necessities, and quantifies the magnitude of this channel on the calorie intake of the poor. Section 7 addresses alternative channels. Section 8 concludes.

2. Background and Theory

A foundational premise of social justice emphasizes the importance of ensuring that every individual has the means to meet their basic needs (Rawls 1971; Sen 1983). This perspective significantly influences the conceptualization of poverty, and relatedly, the discussion on what basic needs truly are. In this section, we substantiate our research

6. Following Hopkins (2024), inequality as relative deprivation assumes asymmetry and cardinality in relative concerns.

7. A notable exception is Brown, Bulte, and Zhang (2011) who look at income rank and consumption in rural China.

question by contextualizing it within the existing literature on basic needs, inequality, and consumption.

A strict definition of basic needs equates them with what Ravallion (2016a) calls the "consumption floor."[8] The consumption floor is often associated with physiological needs, the minimum nutrients needed to survive. This principle aligns with Engel's first law (Engel 1857) and occupies the base of Maslow's "hierarchy of needs" (Maslow 1943; Chai and Moneta 2012). It greatly influences the definition of absolute poverty lines, traditionally determined by the minimum monetary amount required for adequate nutrition.

Nevertheless, the capability approach of Sen (1983) and the abundant literature that follows profess that basic needs are not only physiological (see Ravallion 2016b; Atkinson 2019, for a review). They also include social needs such as social inclusion and self-respect.[9] It is well-documented that poor individuals under malnutrition allocate a non-negligible portion of their income to non-caloric, less necessary items, such as those used during social events (Banerjee and Duflo 2007).

In contrast to absolute poverty, relative poverty measures typically operate under the assumption that the standard of living in a society increases as the society becomes wealthier. This concept underscores the significance of relative concerns (Ravallion and Chen 2011). Research suggests that individuals tend to place greater emphasis on negative comparisons with those who are wealthier than themselves (referred to as "relative deprivation") rather than comparisons with those who have less (Fehr and Schmidt 1999; Clark, Frijters, and Shields 2008). The feeling of relative deprivation, and its behavioral consequences, is closely linked to one of the most widely used measures of inequality: the Gini coefficient (see Clark and D'Ambrosio 2015, for a review). Yitzhaki (1979) was the first to demonstrate that the Gini coefficient can be interpreted as an average measure of relative deprivation in society, based on the work of the sociologist Runciman (1966). Berrebi and Silber (1985) later generalized this finding showing that most commonly used inequality indices can be written as indices of relative deprivation.

The theoretical literature that links inequality and consumption provided different predictions about the effect of inequality at the bottom of the distribution. Inequality has been incorporated directly in consumption utility, either as ordinal preferences (Frank 1985; Hopkins and Kornienko 2004) or as cardinal preferences (Frank, Levine, and Dijk 2014; Hopkins 2024). Models that rely on ordinal preferences predict that an increase in inequality should reduce wasteful consumption, with no or little effect on consumption at the very bottom of the distribution (Hopkins and Kornienko 2004). This

8. Ravallion (2016a) defines the consumption floor as "the lower bound of the support of the distribution of levels of living," which in the demand literature would translate into the sum of basic needs (in expenditure).

9. This idea was already present in Adam Smith (1776), book 5, chapter 2, article 4:"A linen shirt, for example, is, strictly speaking, not a necessary of life. The Greeks and Romans lived, I suppose, very comfortably though they had no linen. But in the present times, through the greater part of Europe, a creditable day-labourer would be ashamed to appear in public without a linen shirt, the want of which would be supposed to denote that disgraceful degree of poverty which, it is presumed, nobody can."

prediction is reversed in models of inequality as relative deprivation, which assume cardinal preferences (Hopkins 2024).[10] In this case, higher inequality leads the poor to increase their consumption on goods consumed by those richer than themselves.

Despite these theoretical predictions, empirical evidence on whether and how inequality as relative deprivation affects the consumption floor is scant. Bertrand and Morse (2016) document that in the United States, the consumption of lower-income groups increases (and savings decline) when exposed to wealthier individuals. However, consumer surveys in developed countries hardly capture consumption at the very bottom of the distribution. And in the absence of price information, one cannot assess whether these effects are due to a shift in basic needs, or to alternative explanations.

In the developing world, Ravallion (2016a) finds no substantial evidence that the (unequal) growth observed in recent decades translated into a significant increase in the consumption floor. Empirical investigations in India even revealed a general decline in nutritional intake across all income levels (Deaton and Drèze 2009), referred to as the "caloric consumption puzzle." In other words, the extremely poor do not eat more today than before. If anything, they eat less.

These seemingly contradictory findings may be reconciled if we consider that inequality influences not just the overall level of consumption at the bottom, but rather its composition. The social inclusion of the relatively deprived may come at a cost: In unequal societies, highly budget-constrained individuals may follow the consumption standards of the rich at the expense of physiological needs. Therefore, we first expect inequality to lead to a calorie loss, which is a simple measure of the overall effect on consumption.

Second, being poor among the affluent may engender a different hierarchy of needs compared to being poor among the poor. Applied to demand theory, the notion of relative deprivation implies that inequality may determine whether a good is perceived as a luxury or a necessity. While the integration of basic needs into demand analysis through the estimation of Engel curves is well-established (Deaton 1986), it remains unexplored how exposure to inequality influences the estimation of income elasticities.[11]

3. Database and Variables

We use all quinquennial rounds of the Indian NSS on Consumption and Expenditure from 1993–1994 to 2009–2010 (50th, 55th, 61st, and 66th rounds). We use two additional quinquennial rounds 1983 (38th) and 1987–1988 (43rd) for the shift-share

10. Hopkins (2024) states that cardinal preferences may be symmetric if utility depends on the average consumption of others as in Bertrand and Morse (2016), or asymmetric if each individual faces a different reference level, which is what we do in this paper.

11. Bramoullé and Ghiglino (2022) explore the theoretical implications of the aversion to lying below the neighbors' average consumption. To our knowledge, this is the only paper that provides a theoretical link between relative consumption and income elasticities.

IV strategy. Each round records the detailed consumption of about 100,000 households. The NSS also provide household survey weights, which we use in all computations and estimations.

3.1.1. Poverty Line. Below poverty line (BPL) households represent a large fraction of the Indian population (36% in 1993–1994, 29% in 2009–2010) and are highly budget constrained, so what they consume is revealing of what constitutes a basic need. We define our sample of BPL households using the official absolute poverty line provided by the Government of India (Planning Commission 2014), which corresponds to the monetary amount required to achieve adequate nutrition. It varies across states, sectors (rural/urban), and rounds, hence, allowing us to compare the consumption choices of households with similar living standards across space and time. In the survey rounds, almost all BPL households are qualified as malnourished (from 94% in 1993–1994 to 98% in 2009–2000; this paper follows the official Indian malnutrition thresholds of 2,100 daily calories in urban areas and 2,400 in rural areas). Online Appendix Table A.1 provides summary statistics of BPL households by survey round for the household variables used in the analysis.

3.1.2. Inequality Exposure. Following the literature discussed in Section 2, we operationalize inequality as a measure of relative deprivation (Yitzhaki 1979; Hopkins 2024). A robust finding of the literature on attitudes to inequality is that comparisons that matter most are highly localized in time and space (Frank, Levine, and Dijk 2014).[12] We, hence, rely on inequality exposure within an administrative district, the smallest geographical unit that can be consistently followed over time. We obtain a total of 413 districts harmonized on the basis of 1987–1988 districts.[13]

We define inequality exposure ρ_{hdt} of a household h in district d in round t with total monthly per capita expenditure m_{hdt}[14] as the sum of all of the gaps between their total expenditure m_{hdt} and the total expenditure m_{ydt} of the set of better-off households $y \in B_{hdt}$. We normalize by the population n_{dt} and the mean total expenditure $\overline{m}_{dt}$ over the entire population so that inequality exposure is bounded between 0 and 1 (Clark and D'Ambrosio 2015):

$$\rho_{hdt} = \sum_{y \in B_{hdt}} \frac{1}{n_{dt}} \frac{(m_{ydt} - m_{hdt})}{\overline{m}_{dt}}. \tag{1}$$

12. Given the possible gap between *actual* and *perceived* inequality (Kuziemko et al. 2015; Bellet et al. 2021), a second advantage of using district-level exposure is that *local* inequality measures are also closer to perceived inequality (Cruces, Perez-Truglia, and Tetaz 2013).

13. We follow the geographical boundaries of the administrative Indian districts as per round 1987–1988 (first to record district information) for all survey rounds, so that a district is the same geographical entity across rounds.

14. Total monthly per capita expenditure, recorded in the surveys, proxies for permanent income. Income is not recorded and is a noisy and imprecise measure of means of living in developing countries, particularly for BPL households who primarily work in the informal sector or home production.

This cardinal measure captures the feeling of deprivation resulting from both the *gap* between the rich and the poor and the *proportion* of richer individuals (Yitzhaki 1979). The mean inequality exposure in our sample of BPL households is 0.44, with an SD of 0.14. Aggregated within a district, the average inequality exposure of BPL household strongly correlates with the district-level Gini coefficient,[15] and, to a lesser extent, with top expenditure shares (Online Appendix Table A.2). The geographical variation in average district inequality exposure is high (Online Appendix Figure A.1).

3.1.3. Expenditure Categories. The survey rounds record monthly household expenditure on detailed items in all consumption categories. The quantities consumed are recorded for all food and drinks, intoxicants, energy use, clothing, and footwear items. The surveys also record home consumption, which is priced at the local market level.

We focus our analysis on the items with recorded quantities, as we use unit values (expenditure divided by quantity) as price proxies. Focusing on these items has another advantage: They correspond to the most frequently purchased items, allowing us to focus on day-to-day budget allocation. The consumption of other categories (durable goods and services) are often single-purchase decisions smoothed over additional periods (e.g., purchase of a fridge, education fees). We follow the demand literature by assuming separability with these categories.

We harmonize the expenditure and quantity of each item following Deaton and Tarozzi (2000).[16] We obtain a total of 167 harmonized items that we aggregate in 12 product categories following the NSS classification (see Online Appendix Table A.3 for a list of items by category). The 12 categories account for the large majority of the expenditure of BPL households, comprising between 80% and 88% of their monthly budget (Online Appendix Table A.5). The budget allocation of BPL households across consumption categories varied over time. The budget share of processed food, meat, dairy, and energy use increased, while the budget share of cereals, highest in the budget of the poor, sharply declined.[17]

3.1.4. Category Prices. We calculate the price index of an expenditure category as the weighted sum of local item prices weighted by their district budget share. We proxy local item prices by their median unit value by village/urban block, which

15. As discussed in Section 2, aggregated across *all* individuals in the district, the measure is exactly equal to the district-level Gini coefficient (Yitzhaki 1979). In India, the Gini index of the average district is 0.27.

16. We systematically draw the quantity and unit value densities for each item by round, and drop the 19 items not recorded in all rounds, or that do not share the same unit (e.g., with multimodal distributions). Online Appendix Table A.4 lists the items we drop by category.

17. Most BPL households in our data record positive expenditure in each category (above 90% of the households), except for dairy products, meat, and intoxicants (Online Appendix Table A.6).

guards against measurement errors and quality choice of the household (Atkin 2013).[18] The weights are the item budget shares of all households at the district-round level to capture heterogeneity in local availability.[19] Prices for food and drink items are normalized in Indian rupees per kilocalorie using the calorie equivalents provided by the NSS.

3.1.5. Nutritional Contents. We use the nutritional data provided by the NSS on the calorie, protein and fat equivalent per quantity of each item. Online Appendix Table A.7 provides the average nutritional content of each category by 2,010 Indian rupee (Rs), as well as the SD of the nutritional content across items within a category. There is substantial variation in how expensive calories, proteins and fats are across categories, with cereals being by far the cheapest source of calories and proteins.[20] Notice that even a category like spices provides some significant amount of calories, proteins and fats per Indian rupee. However, it is unsurprisingly an insignificant contributor to the total nutritional intake, as BPL households spend very little on this category (Rs 0.35 per capita per day, i.e., obtaining on average 12 calories, compared to Rs 3.94 on cereals, that is, 1,246 calories).[21]

4. Engel Curves

4.1. Non-parametric Engel Curves

We first investigate the relationship between inequality exposure and consumption from an Engel curve perspective: We look at how the expenditure on a particular good varies with total expenditure (or income) for poor households more or less exposed to inequality.[22]

We expect two stylized facts consistent with inequality as relative deprivation. First, we test whether the cost of social inclusion translates into a calorie loss.

18. The NSS interview 10 households by village/urban block, and each village/urban block typically has one local market. The median unit value of an item reflects the local market cost. If the item is not consumed at the village/urban block level, we incrementally step one geographical level higher.

19. Formally, the price index $P_{i,vt}$ of a given category i containing n_i items is computed as $P_{i,vt} = \sum_{j=1}^{n_i} w^i_{j,dt} p_{j,vt}$, s.t. $\sum_{j=1}^{n_i} w^i_{j,dt} = 1$, where $p_{j,vt}$ is the median price of item $j = \{1, ..., n_i\}$ in category i in village v in round t, and $w^i_{j,dt}$ is the budget share of item j at the district-round level.

20. The between-category SD in calorie content is more than twice higher than the average within-category SD.

21. We cross-check the validity of the provided nutritional contents for spices. Online Appendix Table A .8 lists the nutritional contents of all eight items of the category spices from the NSS data and from the US Department of Agriculture, or other equivalent sources. The values are remarkably similar, especially given that the exact content of each spice may vary from India to the United States, for example, by adding oil, salt, or other ingredients in the composition of powdered spices.

22. We follow much of the previous literature and use (permanent) "income" and "total expenditure" interchangeably (Heffetz 2011).

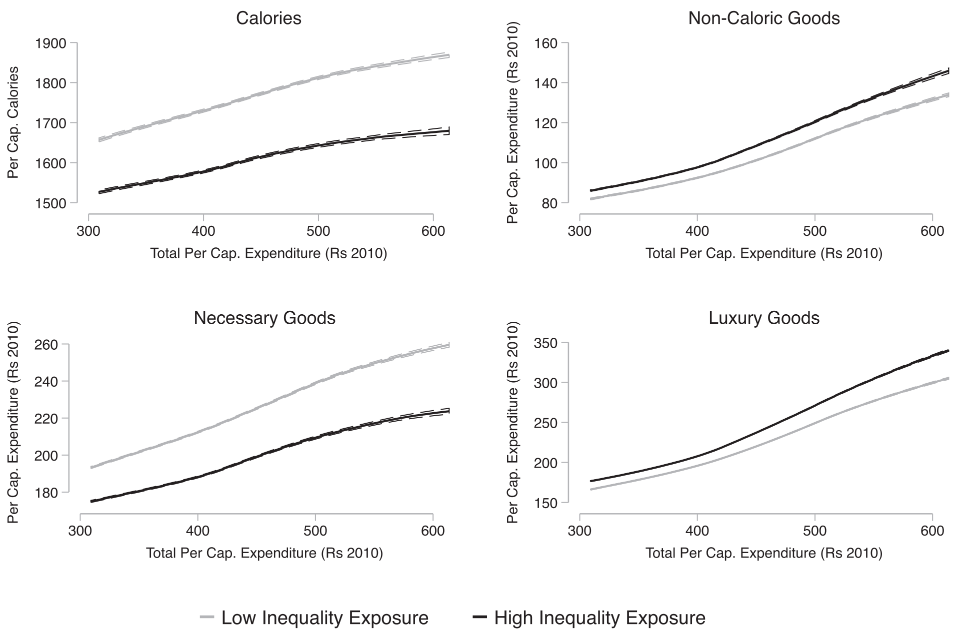

FIGURE 1. Non-parametric Engel curves with variation in inequality exposure. Nonparametric Engel curves for four aggregated consumption categories drawn using the sample of BPL Indian households in the four survey rounds from 1993–1994 to 2009–2010. The Engel curves are obtained by local polynomial smoothing. Households are divided between low (below median) and high (above median) levels of inequality exposure. Monthly per capita expenditure has been adjusted to Indian Rupees (Rs) 2010. The total expenditure cutoffs are the 10th percentile and 90th percentile of the total per capita expenditure distribution. Confidence intervals (CIs; dashed lines) drawn at the 95% level are constructed around the local polynomial regression smoothed estimates.

Unless households are already at the lower bound for metabolic survival, inequality exposure should be associated with a downward shift in the Engel curve of calorie consumption (i.e., a decrease in the intercept, which represents a shift in basic needs given prices). Second, this drop in calorie intake should underline a more general change in the composition of expenditure between necessities and luxuries at the floor. When inequality exposure is higher, we should observe a reallocation of expenditure from necessities to luxuries translated in a shift in their respective Engel curves.

We first find evidence for the first stylized fact, a fall in calorie consumption with inequality exposure. We draw non-parametric Engel curves using consumption data of BPL Indian households in the four rounds from 1993–1994 to 2009–2010. Figure 1 draws two Engel curves in each quadrant by dividing the sample between low (below median) in gray and high (above median) inequality exposure in black.[23] Higher levels

23. The difference in average inequality exposure between both groups corresponds to about 1.5 SD difference in inequality exposure.

of inequality exposure are associated with a clear downward shift in the calorie Engel curve (upper-left quadrant). The loss corresponds to about 150 daily calories per capita at all levels of total expenditure. This drop in calorie intake is particularly striking within a population of vulnerable households already under malnutrition. In contrast, we observe an opposite shift of the Engel curve for non-caloric goods (Figure 1; upper-right quadrant).

We then explore the second stylized fact, a shift from necessities to luxuries with inequality exposure. At this stage, to characterize whether a particular good is a necessity or a luxury we follow Heffetz (2011) and draw non-parametric Engel curves in shares of expenditure on total expenditure for all surveyed households for each of the 12 categories of expenditure (see Online Appendix Figure B.1). Goods such as dairy products, meat, sweets, processed food and drinks, or energy use are luxuries for the entire population. Necessities include cheap sources of calories such as cereals or legumes, but also spices and oil. We aggregate expenditures into luxuries and necessities based on these results. The shift in the Engel curves evidenced in Figure 1 (lower quadrants) reflects what we would expect with a change in needs driven by relative deprivation. Poor households exposed to higher levels of inequality tend to spend less on necessities and more on luxuries.[24] Of course, differences in prices at the local level or in demographics and household characteristics could account for these observed patterns. We examine significant confounding factors and potential endogeneity issues in the following section.

4.2. *Parametric Engel Curves*

4.2.1. Inequality and Calorie Intake. In this section, we assess the robustness of the first stylized fact to confounding factors and endogeneity. We start by constructing the variable $\ln \text{calories}_h$ for every household h, which is the log of total calorie intake per person per day using the reported consumption on the 167 items. Our basic specification regresses this measure on log total per capita expenditure as in a calorie Engel curve estimation (Deaton and Subramanian 1996), with the addition of inequality exposure:

$$\ln \text{calories}_h = \beta \rho_{hdt} + \xi \ln m_h + \Pi X_h + \Gamma_t + \varepsilon_h, \tag{2}$$

where ρ_{hdt} is the household inequality exposure in district d and round t specified in equation (1), and m_h is the household total monthly per capita expenditure. Our parameter of interest, β, is negative if the hypothesis holds, that is, if at any given level of total expenditure, more inequality exposure shifts the calorie Engel curve downward. We normalize ρ_{hdt} using Z-score, so that β captures the change in calorie

24. Online Appendix Figure B.2 shows that this pattern holds when including other consumption categories provided by the NSS: durable goods, toiletries, and services (see Online Appendix Figures B.3 for the Engel curves for those categories using all surveyed households). As mentioned in Section 3, these categories are not included in the analysis due to their purchase frequency (inter-temporal) and non-recorded quantities.

TABLE 1. Inequality exposure and calorie intake.

	Dependent variable: Ln daily per capita calorie intake				
	OLS				2SLS
	(1)	(2)	(3)	(4)	(5)
Inequality exposure (Z-score)	-0.081^{***}	-0.080^{***}	-0.074^{***}	-0.067^{***}	-0.077^{***}
	(0.004)	(0.004)	(0.004)	(0.004)	(0.013)
Observations	90,819	90,819	90,819	90,819	90,819
Mean daily per capita calories	1645.4	1645.4	1645.4	1645.4	1645.4
IV total expenditure	No	No	No	No	Yes
IV inequality exposure	No	No	No	No	Yes
KP F-stat					27.1
Baseline controls:					
Ln total expenditure	Yes	Yes	Yes	Yes	Yes
Round FE	Yes	Yes	Yes	Yes	Yes
Demographics controls	No	Yes	Yes	Yes	Yes
Occupations controls	No	No	Yes	Yes	Yes
Local price index	No	No	No	Yes	Yes

Notes: OLS and 2SLS estimates of the influence of inequality exposure on log daily per capita calorie intake. All regressions control for log total expenditure and round fixed effects. OLS models reported in columns (1–4). Column (2) controls for the demographic structure of the household (the log household size and the proportion of household members by gender and age in five categories, interacted with round fixed effects). Column (3) adds the household occupation (interacted with round fixed effects). Column (4) adds the local price index. Column (5) reports the results of a 2SLS model where both inequality exposure and total household expenditure are instrumented using our shift-share instruments detailed in Section 4.2.2. Robust standard errors clustered at the district-round level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

intake following a 1 SD increase in inequality exposure. Γ_t is a round fixed effect capturing any round-specific trend and ε_h is an error term clustered at the district-round level. Column (1) of Table 1 estimates this simple specification using OLS. Every 1 SD increase in inequality exposure is associated with a highly significant 8% decline in calorie intake. Given the sample mean of 1,645 calories per capita per day, this is equivalent to about 130 fewer calories.

We first test how robust the estimate is to demographics and household characteristics X_h known to affect calorie intake (see Online Appendix Table A.1 for corresponding descriptive statistics). We account for the potential heterogeneity in the demographic structure of the household (Deaton and Subramanian 1996). This includes the log household size as well as the proportion of household members by gender and age in five categories (0–4, 5–9, 10–14, 15–55, and over 55). These variables account for the potential correlation between total expenditure per capita, household composition, and calorie requirement. We allow the coefficients on all household-level controls to differ by survey round. Column (2) shows that the β coefficient remains the same when adding this full list of demographic controls. Following Atkin (2016), we also control for the occupation of the head of the household (interacted with survey round), accounting for the possibility of a correlation between inequality exposure and less physically intensive occupations that require less calories.

The coefficient estimated in column (3) drops slightly but remains highly significant and within the same CI.

The effect may be partly driven by local differences in price levels correlated with inequality. In column (4), we further control for a local stone price index. The stone price index is calculated as the sum over categories of log village/urban block prices weighted by district shares. Controlling for the price level slightly reduces the coefficient from −0.074 to −0.067.

The theoretical literature on inequality as relative deprivation predicts that cardinal (not ordinal) preferences may increase wasteful consumption at the bottom of the distribution (Hopkins 2024). Inequality exposure as relative deprivation is a cardinal measure of preferences, which depends on the relative income of richer others in the district. Online Appendix Table C.2 confirms that adding a strictly ordinal measure as control (the fraction of richer others) does not affect the estimate on relative deprivation (column 2). Moreover, using the Gini coefficient (or average relative deprivation in the district), which unlike the household-level measure does not depend on own income rank but captures the average level of relative deprivation in the district (a cardinal measure), also leads to a negative and significant effect (Online Appendix Table C.2; column 3).

Finally, we also estimate regression 2 in levels rather than the standard log–log Engel curve specification. This is consistent with the linearity of the non-parametric Engel curves for the BPL households documented earlier, and reflects the assumption of quasi-homotheticity of our demand system in Section 5. We find similar effect sizes when using this alternative Engel curve specification. The estimated calorie loss following a 1 SD in inequality ranges between 134 calories (95% CIs: −146.2, −122.5) and 113 calories (CIs: −125.1, −101) per capita per day (Online Appendix Table C.1).

4.2.2. Endogeneity Concerns and Instrumentation. The OLS estimation assumes that cross-sectional differences in inequality exposure are exogenous. This may be a reasonable assumption given that a significant portion of the local variation in inequality is structural, deeply rooted in the historical influence of caste and religious affiliations. Adding baseline household-level controls known to affect calorie intake, or adjusting for local differences in prices, has little impact on the documented effect (Table 1). However, there exists three potential threats to a causal interpretation.

First, there is a possibility of reverse causation between local inequality exposure and household preferences for particular products. If some households have a taste for luxuries or status goods, they may consume less calories, hence, depleting their working capacity and becoming poorer both in absolute and relative terms (Dasgupta and Ray 1986). This would bias the estimate downward.[25]

Second, omitted current district or household variables may affect inequality locally while being also correlated to consumption patterns. For instance, if richer

25. Conversely, if status goods have instrumental values, those same households would obtain a higher bargaining power on labor markets (and higher income), leading to an upward bias on the estimate (Ball et al. 2001).

people possess local industries of luxuries and the local population develops a higher taste for these goods due to advertising, then the consumption of these goods would generate profit that increases local inequality. The local availability of luxury goods may also attract wealthier migrants while providing an easier access to these goods for the relatively deprived. These effects would bias our estimate downward. Finally, classical measurement errors in recorded expenditures may lead to biased estimates toward zero.

We construct shift-share instrumental variables for total expenditure and inequality that (i) are not a reflection of household preferences and (ii) do not originate from current district characteristics that determine inequality levels locally. We rely on two features of the Indian context. First, occupational choice is still largely determined by traditional occupations rigidly tied to the caste and religious affiliations given at birth (Atkin, Colson-Sihra, and Shayo 2021; Cassan, Keniston, and Kleineberg 2021; Oh 2023).[26] Second, India experienced liberalization and economic reforms in the early 1990s that drastically changed occupational returns nationally. The period that followed was associated with a sudden and drastic increase in income inequalities driven by the upper part of the income distribution (Chancel and Piketty 2019).[27]

Our IVs combine the national growth in occupational returns between round 1987–1988 (pre-liberalization) and each subsequent survey round (post-liberalization) with the local representation of occupations within each caste/religion (Atkin, Colson-Sihra, and Shayo 2021). We rely on 12 occupation codes (one digit),[28] and 8 caste or religious identities,[29] to obtain the predicted total expenditure m_{cdt} for a caste/religion c in district d and round t:

$$m_{cdt} = \sum_{o} m_{od^{-}t} \theta_{ocdt_0}, \tag{3}$$

where $m_{od^{-}t}$ is the national occupational return of occupation o in round t leaving out own district d, and θ_{odct_0} is the occupation share by caste-district in the pre-liberalization survey round t_0 (1987–1988).[30] This shift-share instrument captures the

26. Caste is determined at birth. Religious conversions are exceedingly rare in India, with 98% of Indian adults still identifying with the religion in which they were raised in 2020 (Sahgal and Evans 2021). It is also legally restricted (Jenkins 2008).

27. The bottom 50% income share fell from a stable 22% of national income between 1960 and 1990 to 15% by 2010 (Online Appendix Figure D.1.1). The causal impact of those reforms (removing licensing and trade restrictions) on inequality is well-documented (Aghion et al. 2008; Topalova 2010).

28. Occupation codes correspond to a one-digit classification, with agriculture—the largest occupation category in population share—being additionally divided into three broad subcategories (agricultural laborers; farmers and agricultural managers; forestry hunters and fishermen).

29. Scheduled Tribes, Hindu Scheduled Castes, Hindu Other Castes, Muslims, Christians, Sikhs, Jains, and Buddhists (see Online Appendix Table A.1 for summary statistics).

30. To reduce endogeneity concerns, we follow Adao, Kolesár, and Morales (2019) and Atkin, Colson-Sihra, and Shayo (2021) and leave out own district from the computation of national growth in occupational returns.

change in total expenditure driven by national changes in returns to occupations in which a caste/religion has a historical prominence in a district.

We construct the predicted inequality exposure ρ_{cdt} for caste/religion c in district d in round t as follows:

$$\rho_{cdt} = \sum_{c' \in B_{cdt}} s_{c'dt_0} \frac{(m_{c'dt} - m_{cdt})}{\overline{m_{dt}}} \quad \text{where } m_{c'dt} > m_{cdt}, \tag{4}$$

where B_{cdt} is the set of castes/religions with higher predicted return than caste/religion c in district d and round t; $s_{c'dt_0}$ is the pre-liberalization (1987–1988) share of caste/religion c' in district d; m_{cdt} ($m_{c'dt}$) is the predicted total expenditure of caste c (other castes c') in district d in round t using equation (3); and $\overline{m_{dt}}$ is the predicted mean district total expenditure based on initial district population.[31]

The validity of shift-share IVs follows either from the quasi-random assignment of shocks across areas (i.e., nationwide shocks in the relative returns to occupations in the period following 1987–1988; Borusyak, Hull, and Jaravel 2021), or from the exogeneity of exposure shares (i.e., occupation shares within caste/religion across districts pre-liberalization; Goldsmith-Pinkham, Sorkin, and Swift 2020). A possible threat to the exogenous assignment of shocks is the presence of systematic pre–post liberalization trends in nationwide growth to occupational returns. We find no evidence of such pre-trends, whether we use 1-digit or 2-digit occupation codes (Online Appendix Figure D.1.2). Exposure to shocks should also not be systematically driven by the same occupation within a given district-caste group. This issue may arise if the distribution of occupations pre-liberalization is not sufficiently dispersed across districts. We find considerable geographical dispersion in the historical exposure to particular occupations prior to the 1991 liberalization (Online Appendix Figure D.1.3). Importantly, this historical exposure varies significantly both between and within caste groups, as illustrated in Online Appendix Figure D.1.4 for agricultural laborers and sales workers.

The 2SLS model relies on our IV m_{cdt} and ρ_{cdt} for total expenditure and inequality exposure, respectively. The Kleibergen–Paap (KP) F-statistic for the joint significance of all instruments is 27.1. Online Appendix Figure D.1.5 (left panel) plots the residualized (bin) scatter plot for the first-stage regression of log total per capita expenditures, after partialing out round fixed effects, predicted inequality exposure and baseline controls. The relationship is strong. Every 1% increase in predicted income leads to a 0.23% increase in total per capita expenditure. We reproduce this exercise for predicted inequality exposure (Online Appendix Figure D.1.5; right panel). Here, too the relationship is strong, with a point estimate for the first-stage of 0.72.

31. Boustan et al. (2013) instrument the local Gini index using a similar—although less exogenous—approach. They rely on the local initial income distribution across 15 income bins and nationwide patterns of income growth. We additionally exploit the pre- (vs. post-) liberalization reform period together with the constraint on occupation choices determined by caste or religion at birth.

Column (5) of Table 1 reports the second stage coefficient from the 2SLS model with baseline controls. The IV estimate of −0.077 is very close to the non-instrumented OLS estimate. In terms of calorie loss, a one standard-deviation increase in inequality exposure is equivalent to roughly 127 fewer calories consumed per day.[32] We test the robustness of our instrument strategy in two additional IV specifications. First, we estimate a more demanding specification by adding round-caste/religion and district-caste/religion fixed effects (Online Appendix Table C.3; column 2). Here, identification relies exclusively on *within* district-caste/religion *changes* in occupational returns post-liberalization. It is not driven by round-specific (e.g., Hindu–Muslim conflict) or district-specific (e.g., Gujarati Jain network in diamond trade) caste/religious shocks that may correlate with preferences. Second, we use the more granular 2-digit occupation codes rather than single-digit occupations to construct our instruments (Online Appendix Table C.3; column 3). Finally, we also replicate the analysis on the full sample (Online Appendix Table C.3; column 4). We find qualitatively and quantitatively similar estimates in all three specifications.

We perform a final test to the validity of our instruments that uses the difference between historical occupational shares in a district, on which both instruments are based, and current ones. For the district-religion groups for which the historical occupational shares and current ones are very distant, the instrument should not be predictive of local exposure to inequality, and, hence, would not affect calorie intake through this channel. We compute an aggregate measure of distance for each caste-district between their occupation shares in the pre-liberalization (1987–1988) round and in round t.[33] We divide observations in quintiles of distance, such that the first quintile of distance has the least difference in occupational shares; hence, the instrument should be most predictive of local inequality. Conversely, the instrument for the sub-sample of the last quintile of distance is close to a "placebo," in the sense that it should be least predictive of local inequality. Compared to the baseline estimates reproduced in column (1) of Online Appendix Table C.4, the first quintile of distance between occupational shares in column 2 indeed displays a stronger reduced-form effect of instrumented inequality exposure on calorie intake (−0.054 compared to −0.035), and one and a half times the 2SLS estimates (−0.116 compared to −0.077). The second to fourth quintiles aggregated in column 3 display results close to the baseline column 1. Finally, the last quintile in column 5, the "placebo" quintile, displays a much weaker point estimate (−0.017) significant at only the 10% level, while the 2SLS estimate is not significant with no first stage (KP F-statistics of 0.9).[34]

4.2.3. Unpacking Relative Deprivation and the Calorie Loss. This section presents additional evidence that aligns with relative deprivation theory. First, we unpack the

32. The estimate in level reaches 132 calories per day (95% CIs: –177.5, –86) (Online Appendix Table C.1).

33. As a measure of differences, we use the Manhattan distance, which is the sum of the absolute values of differences in shares. The results are stable when using the Euclidean distance instead.

34. The instruments may still have limited predictive power for the last quintile, due to a weak correlation between historical and current shares, combined with potentially large changes in occupational returns over time.

relative deprivation effect and test whether the calorie loss is driven by upward or downward comparisons. Second, we test if the second stylized fact, the fact that the relatively deprived increase their consumption of goods commonly consumed by the rich, is robust to our controls and instrumentation strategy.

We construct three alternative measures of inequality exposure. First, instead of upward comparisons, we define downward-looking inequality exposure as the sum of all of the gaps between a household's total expenditure m_{hdt} and the total expenditure m_{ydt} of the set of worse-off households $y \in W_{hdt}$.[35] Next, we construct two separate upward-looking measures dividing better-off households into two segments: the middle-class, earning up to 1.5 times more than the poverty line, and the upper-class, earning beyond this threshold. This division splits the sample of non-poor households in two groups of relatively equal size. We derive the corresponding inequality IVs for these three measures.

Column (2) of Online Appendix Table C.5 shows that the calorie loss is driven by upward comparisons with richer households and not downward comparisons among the poor. Additionally, exposure to a richer middle class has a stronger impact on the calorie loss than exposure to the upper class (columns 3–4). Unlike peer effects, the feeling of relative deprivation is only upward looking. It is also stronger when the poor are exposed to a group of richer others that they are more likely to encounter or compare with in their daily lives.[36]

The drop in calorie consumption underlies a more fundamental change in the intra-temporal expenditure allocation of poor households. The 2SLS estimates in Table 2 confirms the results from the non-parametric Engel curves. First, poor households under malnutrition spend more on non-caloric items when exposed to higher levels of inequality. Second, using the same classification as in Section 4.1, we test how inequality affects the allocation of expenditure between necessities and luxuries. Consistent with relative deprivation theory, the negative estimate on calorie intake underlies a more general expenditure reallocation toward luxuries away from necessities. A 1 SD increase in exposure leads poor households to reduce their consumption of basic necessities by 7.5% (column 2) and increase their expenses on luxuries by 4.5% (column 3). Results are robust to the inclusion of luxury and necessary expenditure with non-recorded quantities (Online Appendix Table C.6).

5. A Model of Relative Needs

In this section, we introduce a demand system that brings three main advantages. First, we wish to estimate the impact of inequality exposure on the needs of the poor separately from the supply and budget constraints. A structural model allows us to

35. This measure is also called "relative satisfaction" in opposition to relative deprivation (Yitzhaki 1979).

36. This is consistent with the fourth axiom of relative deprivation from Runciman (1966), which is that a person feels relatively deprived of X when "they see it as feasible that they should have X."

TABLE 2. Inequality exposure and other expenditure categories.

	Dependent variable: Ln monthly per capita expenditure		
	Non-caloric goods (1)	Necessities (2)	Luxuries (3)
Inequality exposure (Z-score)	0.076*** (0.023)	−0.075*** (0.017)	0.045*** (0.014)
Observations	90,730	90,752	90,811
Sample mean (daily Rs)	111.6	212.2	240.8
IV total expenditure	Yes	Yes	Yes
IV inequality exposure	Yes	Yes	Yes
Baseline controls	Yes	Yes	Yes

Notes: 2SLS estimates of the influence of inequality exposure on log monthly per capita expenditure in each group. The dependent variable is log per capita expenditure on non-caloric goods in column (1), on necessities (cereals, oil, legumes, and spices) in column (2), and on luxuries (dairy, meat, energy use, vegetables, clothing, processed food and drinks, sweets, and intoxicants) in column (3). Necessities and luxuries are classified using non-parametric Engel curves from Online Appendix Figure B.1. All regressions control for the log of total expenditure, round fixed effects and the full list of baseline controls. Inequality exposure and total expenditure are instrumented using our shift-share instruments detailed in Section 4.2.2. Robust standard errors clustered at the district-round level are reported in parentheses $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

specify and estimate needs separately. Second, the demand system accounts for the simultaneity of the budget allocation decision among 12 disaggregated consumption categories. Finally, the demand system allows us to estimate the impact of inequality on the hierarchy of needs (or income elasticities), and run counterfactual scenarios.

We rely on a Stone–Geary utility function, in which consumers value a necessary quantity of each category i. Given income m and a price vector P, we assume that each consumer maximizes their utility $U(Q)$ from consuming a vector Q of quantities. Formally, the consumer maximizes their utility subject to the budget constraint[37]:

$$\begin{aligned} U(Q) &= \sum_i \beta_i \ln(q_i - \gamma_i), \\ \text{with } &\sum_i \beta_i = 1, \quad q_i - \gamma_i > 0, \\ \text{subj. to: } &\sum_i p_i q_i = m. \end{aligned} \tag{5}$$

The preference parameter γ_i is the necessary quantity for each good i. Intuitively, it represents a reference level, or basic need, for each good. This need can be driven by physiological subsistence (e.g., calories), but also by cultural norms or reference levels. It captures the "consumption floor" (Ravallion 2016a) for each good in a population.

Above the total necessary level of expenditure, $\sum_i p_i \gamma_i$, the individual allocates income m according to taste parameters β_i. Two standard restrictions are imposed on the parameters. First, the additivity restriction $\sum_i \beta_i = 1$. Second, the system is not

37. To simplify notation, we use generic q_i, p_i, and so on, with the understanding that the problem is household h specific.

defined if the individual does not consume the necessary amount of each good, that is, if $q_i - \gamma_i \leq 0$. The individual needs to afford the necessary quantity of not just a few, but all goods.

We add to this framework by assuming that consumer needs γ_i depend on household inequality exposure ρ given by equation (1). This assumption translates into inequality exposure shifting the Engel curve of each good, rather than affecting its slope. It is empirically consistent with the pattern that emerges from the non-parametric Engel curves (Figure 1). We consider the following additive specification[38]:

$$\gamma_i = \tau_i + \nu_i \rho. \tag{6}$$

Inequality exposure translates into the commodity space by affecting the need for each good i differently. The parameter ν_i captures this differential effect across goods (i.e., whether the Engel curves shift upward or downward).

Solving the consumer problem (5) with the formula for consumer needs in equation (6), we obtain the Marshallian demand function in expenditure x_i for each good i:

$$x_i = q_i p_i = (\tau_i + \nu_i \rho) p_i + \beta_i \left(m - \sum_j (\tau_j + \nu_j \rho) p_j \right). \tag{7}$$

The set of expenditure functions constitutes a demand system called the linear expenditure system (LES) in the demand literature. The LES belongs to a family of demand systems generated by the Gorman polar form (Blundell and Ray 1984). This family of demand systems, unlike the almost ideal demand system (AIDS, Deaton and Muellbauer 1980), allows us to directly estimate needs. Compared to the AIDS, however, the LES has two restrictions for its application to data: First, quasi-homothetic preferences mean that the Engel curves are assumed to be linear. This assumption fits well the demand patterns of poor Indian households in our specific context (see Figure 1), but should be confirmed in other applications.

Second, the LES assumes separable needs, that is, does not flexibly account for substitution patterns across expenditure categories. We can relax this assumption to assess how constraining it is in our setting. A generalization of this demand system, labeled linear preference system, provides a flexible functional form in the price space (Blundell and Ray 1984). We develop this generalization in Online Appendix Section E and show that, although less tractable, it preserves our testable prediction. We also estimate this demand system and compare the results of both structural estimations in the next section.

5.1.4. Testable Prediction: Relative Necessities. A testable prediction of our model is that inequality exposure increases the need for goods that bring social inclusion. We

38. Pollak (1970, 1976) proceeds to a similar linear decomposition to introduce habit formation or demographic components in this demand system. Other works introduce interdependent preferences, or peer effects, through the same channel (Pollak 1976; Bowles and Park 2005; Lewbel et al. 2022).

call these goods *relative necessities*. Given the budget constraint and assumptions, an increase in the demand for certain goods necessarily implies a substitution with other goods.[39] We found preliminary evidence for substitution in Section 4, but the model provides an exact expression for the reallocation of expenditure across goods driven by inequality. Differentiating the demand equation (7) with respect to ρ, we obtain the difference in expenditure driven by a marginal change in inequality exposure for each good i:

$$\partial x_i = \left[\nu_i p_i - \beta_i \sum_j \nu_j p_j \right] \partial \rho. \tag{8}$$

Good i is a relative necessity if and only if

$$\nu_i p_i > \frac{\beta_i}{1-\beta_i} \sum_{j \neq i} \nu_j p_j.$$

Intuitively, if good i is a relative necessity, inequality exposure increases the need for good i ($\nu_i p_i$) relatively more than for the other goods ($\sum_{j \neq i} \nu_j p_j$).

In the context of dire poverty, the expenditure reallocation may be at the expense of other needs such as physiological sustenance. In effect, inequality exposure can alter the hierarchy of needs. This expression provides a straightforward way to quantify such consequences.

A corollary of this testable prediction is that inequality exposure shifts down the income elasticity of relative necessities, rendering them relatively more necessary. Formally, from the demand equation (7), we derive the income elasticity ξ_i for each good i:

$$\xi_i = \frac{1}{1 + (\tau_i + \nu_i \rho) \frac{1}{\beta_i} \frac{p_i}{m} - \sum_j (\tau_j + \nu_j \rho) \frac{p_j}{m}}. \tag{9}$$

Differentiating equation (9) with respect to ρ, we obtain the change in income elasticity driven by a marginal change in inequality exposure:

$$\partial \xi_i = \frac{-\frac{1-\beta_i}{\beta_i} \nu_i \frac{p_i}{m} + \sum_{j \neq i} \nu_j \frac{p_j}{m}}{\left[1 + (\tau_i + \nu_i \rho) \frac{1}{\beta_i} \frac{p_i}{m} - \sum_j (\tau_j + \nu_j \rho) \frac{p_j}{m} \right]^2} \partial \rho. \tag{10}$$

Our model of relative needs predicts that inequality exposure shifts the hierarchy of needs: When poor households are exposed to richer households, social inclusion needs

39. By assumption, inequality exposure does not increase the aggregate expenditure in net, but reallocates expenditure across goods. The sum of the difference in expenditure with inequality exposure across all goods equals 0 ($\sum_i (\nu_i \rho p_i - \beta_i \sum_j \nu_j \rho p_j) = 0$) due to the additivity restriction $\sum_i \beta_i = 1$. This assumption may not hold if inequality affects income levels too. We find that local inequality is not correlated with total expenditure once it is instrumented (Online Appendix Table D.1.1). Online Appendix Table A.1 also shows very limited change in average total expenditure of BPL households over the period, consistent with what Ravallion (2016a) underlines across countries.

become relatively more important than other basic needs. Given that luxury goods (income elasticity above 1) are more likely to be used for social inclusion motives than necessities (relatively more consumed by the poor), an empirical implication of inequality as relative deprivation is that absolute luxuries are also relative necessities.

6. Structural Estimation

6.1. Specification

Using the demand system developed in Section 5, we estimate the monthly per capita demand of BPL households in the four survey rounds from 1993–1994 to 2009–2010. From the expenditure functions in equation (7), we obtain a system of 12 demand equations for per capita expenditure $x_{i,ht}$ on category i of household h in round t (in village v in district d):

$$\left\{\begin{aligned} x_{1,ht} &= (\tau_1 + \nu_1 \rho_{hdt} + \delta_{1,t}) P_{1,vt} + \beta_1 m_{ht} \\ &\quad - \beta_1 \left[\sum\nolimits_j (\tau_j + \nu_j \rho_{hdt} + \delta_{j,t}) P_{j,vt}\right] + \varepsilon_{1,ht} \\ x_{2,ht} &= (\tau_2 + \nu_2 \rho_{hdt} + \delta_{2,t}) P_{2,vt} + \beta_2 m_{ht} \\ &\quad - \beta_2 \left[\sum\nolimits_j (\tau_j + \nu_j \rho_{hdt} + \delta_{j,t}) P_{j,vt}\right] + \varepsilon_{2,ht} \\ &\vdots \\ x_{12,ht} &= (\tau_{12} + \nu_{12} \rho_{hdt} + \delta_{12,t}) P_{12,vt} + \beta_{12} m_{ht} \\ &\quad - \beta_{12} \left[\sum\nolimits_j (\tau_j + \nu_j \rho_{hdt} + \delta_{j,t}) P_{j,vt}\right] + \varepsilon_{12,ht} \end{aligned}\right. , \tag{11}$$

where m_{ht} is the total monthly per capita expenditure of the household on the 12 categories; $\tau_i + \nu_i \rho_{ht} + \delta_{i,t}$ is the necessary quantity of category i; τ_i is the basic component; and $\nu_i \rho_{hdt}$ is the relative component driven by inequality exposure ρ_{hdt} of household h in district d in round t (see equation (1)). The demand equations of the system (11) suffer from the same endogeneity concerns detailed in Section 4.2.2. We hence instrument m_{ht} and ρ_{hdt} by using the shift-share instruments for total expenditure and inequality exposure. Our specification also includes category-round intercepts $\delta_{i,t}$ to accommodate round-specific effects on expenditure.[40] We mean-center inequality exposure, so that $\tau_i = \gamma_i$ (the basic need, or consumption floor) for the representative household in our sample. Error terms $\varepsilon_{i,ht}$ are clustered at the district-round level. $P_{i,vt}$ is the price of category i in village/urban block v in round t (see Section 3 for its construction).

40. It would be unfeasible to estimate the demand system with additional variables given its nonlinear nature and high number of parameters (6 × 12 categories as it is), and a sample with 91,412 observations. Reassuringly, Section 4.2 shows that the effect of inequality exposure on calorie loss and aggregated expenditures is qualitatively and quantitatively similar with additional controls and fixed effects.

We use the multiple-equation generalized method of moments estimator to account for the nonlinearity of the demand system (11) and the simultaneity of the budget allocation decision among the 12 categories. The estimated parameters of the instrumented system of demand equations (11) are reported in Table 3.

6.1.1. Price Instruments. Concerned about a potential source of endogeneity from prices, we instrument prices using the price in a village in another district in the same state (Atkin 2013).[41] Inequality could increase the local taste for relative necessities, and if competition is imperfect, we may expect higher local market prices relative to other categories. Sellers may also charge higher prices when buyers have a preference for status (Bagwell and Bernheim 1996). Both effects would attenuate the demand for relative necessities in places more exposed to inequality. The reverse could be true if a higher demand for such goods spurs innovation and economies of scale, hence, lowering the price relative to other goods (Jaravel 2019; Atkin et al. 2020). This effect is of greater concern, as it would lead to overestimating the relative necessary quantity for such categories in places more exposed to inequality. Preliminary evidence suggests, reassuringly, that the first bias seems to dominate the second and that it vanishes once we use our instrument for inequality exposure, and a fortiori with both price and inequality instruments (Online Appendix Figure D.2.1).

There may also be adjustments in the quality of the products being purchased that co-vary with local inequality exposure. The price index of each expenditure category captures part of this effect, as long as quality differences are reflected in differences in local prices and availability. We can assess, however, the importance of this quality-based channel looking more closely at the composition of items *within* each category. To test whether inequality drives the item composition more toward higher quality items than toward lower quality items, we compute the median price of each item within a product category in each district, and regress the 90/10 percentile ratio of (weighted by district-round budget shares) median item prices on district average inequality exposure. Online Appendix Figure D.2.2 confirms inequality is correlated with the availability of highly priced (vs. lower priced) items, but also shows that the correlation entirely vanishes once we use our instrumented measure of inequality exposure.

6.2. The Consumption Floor: Basic Necessary Quantities

The basic necessary quantities τ_i reflect the basic needs of poor households in India. Our empirical results are in line with this interpretation: First, the estimated basic

41. This price instrument captures supply costs uncorrelated with local permanent characteristics. For the instrument to be valid, we require that supply shocks are correlated spatially within states, but idiosyncratic village characteristics across districts are not. Atkin (2013) provides supporting evidence of these assumptions for food products in India for nearby villages in the same district. Using prices in a different district within the same State reinforces the validity of the instrument, especially given that our measure of inequality exposure is calculated within a district.

TABLE 3. Estimated parameters of the demand system with inequality exposure.

	Dependent variable: per capita expenditure on											
	Cereals	Clothing	Dairy	Energy	Intoxicants	Legumes	Meat	Oil	Processed	Spices	Sweets	Vegetables
Taste parameter β_i	0.074	0.122	0.137	0.152	0.042	0.062	0.080	0.058	0.032	0.040	0.072	0.130
	(0.008)	(0.003)	(0.007)	(.)	(0.001)	(0.004)	(0.004)	(0.003)	(0.002)	(0.001)	(0.002)	(0.004)
Basic necessary quantity τ_i	26.474	0.120	−0.157	3.169	−0.028	0.492	−0.065	1.532	0.076	0.062	−0.029	0.378
	(0.654)	(0.010)	(0.116)	(0.235)	(0.022)	(0.086)	(0.016)	(0.134)	(0.011)	(0.008)	(0.016)	(0.037)
Relative component ν_i	−60.294	−0.324	1.912	−0.352	0.148	−0.237	−0.079	−1.274	0.118	0.160	−0.136	−1.165
	(2.277)	(0.069)	(0.537)	(1.279)	(0.137)	(0.317)	(0.070)	(0.449)	(0.038)	(0.056)	(0.126)	(0.198)
Round 1993–1994	4.134	−0.070	−0.343	−5.345	0.151	−0.270	−0.075	−0.968	−0.099	−0.021	−0.071	−0.316
	(0.453)	(0.011)	(0.090)	(0.272)	(0.030)	(0.069)	(0.011)	(0.083)	(0.009)	(0.009)	(0.020)	(0.032)
Round 1999–2000	1.822	−0.062	−0.298	−3.113	0.085	0.101	−0.017	−0.718	−0.094	0.029	−0.101	−0.156
	(0.434)	(0.011)	(0.085)	(0.230)	(0.023)	(0.057)	(0.011)	(0.085)	(0.009)	(0.009)	(0.020)	(0.032)
Round 2004–2005	1.837	−0.067	−0.679	−2.210	0.012	−0.427	−0.073	−0.463	−0.112	−0.040	−0.138	−0.257
	(0.477)	(0.010)	(0.088)	(0.226)	(0.018)	(0.072)	(0.011)	(0.081)	(0.008)	(0.008)	(0.019)	(0.031)
Observations	91,412	91,412	91,412	91,412	91,412	91,412	91,412	91,412	91,412	91,412	91,412	91,412

Notes: Dependent variable is the monthly household per capita expenditure on each 12 categories. Each column is one of the simultaneously estimated 12 demand equations. β_i is the taste parameter, τ_i is the basic component of necessary quantities, and ν_i is the relative component of necessary quantities driven by inequality exposure. Monthly household per capita total expenditure is instrumented by national returns to the initial (1987–1988) district occupational mix of own caste/religion. Inequality exposure is instrumented by predicted inequality exposure based on the instrument for total expenditure. Both instruments are detailed in Section 4.2.2. Prices are instrumented by the price in a village in another district in the same state. The estimation includes round-specific effects, with the 2009–2010 round as baseline. No standard error is estimated for the parameter β_i for energy to fulfill the additivity restriction $\sum_j \beta_j = 1$. Robust standard errors clustered at the district-round level are in parentheses. Regressions are weighted by survey population weights.

necessary quantities τ_i in Table 3 are positive or insignificant, except for meat.[42] Second, cereals are by far the most important estimated basic necessary quantity. The demand system estimates that poor households require 882 calories per capita per day from cereals.[43] This amount is affordable to poor households: It amounts to Rs 84 of cereals per capita per month.[44] To give an order of magnitude, this monthly spending for one person can be obtained in less than 1 day of work under the minimum daily wage for rural workers in 2009.[45] Third, other intuitively necessary categories such as energy use, vegetables, legumes, and cooking oil have a positive, although smaller, estimated basic necessary quantity.

We find that the total basic calorie intake predicted by our estimation is in the range of metabolic survival. We estimate this basic calorie intake at 959 calories per capita per day.[46] This amount is close to the lower bound for metabolic survival given in the medical literature (800 calories per day),[47] but less than half the official Indian threshold for malnutrition (2,100 calories per day in urban areas and 2,400 in rural areas). This result suggests that the estimation of basic necessary quantities captures what we expect as physiological need. It also underlines the difference between starvation and malnutrition thresholds, and the amount of choice one might exert in between.

Another way to gauge at the ability of our demand system to capture the "consumption floor" is to calculate the price of the necessary basket and compare it to the actual total expenditure in our data. We find that the necessary basket is affordable for virtually all households in our sample. Online Appendix Figure F.1 draws monthly per capita expenditure on the necessary basket (basic necessary quantities multiplied by local prices) as a percentage of total monthly per capita expenditure for all BPL households. The vast majority of our sample is well above the necessary level of expenditure. The total necessary expenditure still accounts for a substantial share of the budget of poor households in India, with a peak at around 30% of their monthly per capita budget.

42. The negative parameter τ_i on a category allow the system to be defined at zero. Intuitively, this category is not necessary for strict survival, and is only consumed once the individual meets the necessary expenditures in all other categories, plus an additional level of income.

43. Recall that we normalize prices in kilocalories for caloric categories, hence, estimated quantities are expressed in kilocalories for these categories. From Table 3, $\tau_{\text{Cereals}} = 26.474$ kilocalories, that is 26,474 calories per capita per month, or 882 per day.

44. On average, 1 Indian rupee spent on cereals is equivalent to 315.45 calories (see Online Appendix Table A.7). Hence, household needs to spend approximately Rs 84 per capita per month to reach the estimated necessary quantity of cereals.

45. The minimum daily wage is fixed by the Mahatma Gandhi National Rural Employment Guarantee Act. In 2009, most states fixed it at Rs 100 per day (about \$2.20).

46. From Table 3, the sum of all τ_i for caloric (food and drinks) categories is 28.76 kilocalories per capita per month. We multiply by 1,000 and divide by 30 to obtain 959 daily per capita calories.

47. This threshold has been used in clinical "starvation" diets in the study of obesity (Ball, Canary, and Kyle 1970; Willms, Ebert, and Creutzfeldt 1978).

The estimated basic necessary quantities are relatively similar across rounds. Online Appendix Figure F.2 compares the basic necessary quantities across rounds relative to the baseline (2009–2010). The basic necessary quantities for categories with a small or insignificant τ_i are similarly small or insignificant for earlier rounds, except for dairy products. For significantly positive necessary quantities, vegetables, cooking oils and energy use show an increase over time, while the necessary quantity for cereals decreases over time. This finding is consistent with the increase in inequality over time (whose contribution we later quantify), but also alternative hypotheses such as an improving disease environment (Deaton and Drèze 2009; Duh and Spears 2017).

These results taken together are not only reassuring for the interpretation of our estimated parameters, but they also underline the capacity of the demand system to capture needs effectively. This interesting property of this family of demand systems has been largely underexploited in the literature, and gives the possibility to further explore changes in basic needs across time, geography, or population.

6.3. Relative Necessities and the Hierarchy of Needs

The testable prediction of our model is that inequality exposure affects demand and income elasticities through the needs of the poor. The estimation first confirms that inequality exposure induces substantial expenditure reallocation away from physiological needs. We calculate the expenditure reallocation driven by 1 SD (0.14) increase in inequality exposure (compared to the representative household) using equation (8) combined with the estimated parameters of our baseline specification. Table 4 shows, in the first column, the consumption floor in expenditure (basic necessary quantity τ_i multiplied by average price). The second column presents the expenditure reallocation for each category per capita per month predicted by an additional SD of inequality exposure. The two categories from which people spend less when exposed to inequality (cereals and vegetables) are also among the categories whose consumption floor is the highest.

Conversely, our estimation identifies categories predominantly associated with social inclusion as relative necessities. The highest increase in expenditure goes to dairy products, which account for about a third of the decrease in expenditure on cereals and vegetables. They are followed by energy use, spices, processed food and drinks, meat, and sweets. These categories are particularly meaningful in social interactions in the Indian context. Dairy products are deeply linked to the food practices and reverence toward the cow of the high castes, which are over-represented in the upper strata of the income distribution (Munshi 2019). Their status as relative necessities could be a sign of "sanskritization" (Srinivas 1956), that is, the mimicry of high-caste practices for social inclusion motives. Energy use captures the flow of usage of durable goods like home appliances, which have high levels of social visibility (Heffetz 2011; Khamis, Prakash, and Siddique 2012), particularly in poor neighborhoods where houses are generally open to the street. Pallavi (2016) reports that spices are "the hallmarks of rich people's food" and

TABLE 4. Reallocation of expenditure and calories with inequality exposure.

	Consumption floor	Reallocation with +1SD inequality exposure	
	Monthly expenditure (Rs)	Monthly expenditure (Rs)	Daily calories
Cereals	71.58	−20.92	−219.95
Vegetables	9.93	−0.76	−1.67
Clothing	7.97	0.32	–
Oil	8.05	0.64	2.68
Intoxicants	−0.32	1.39	–
Legumes	4.45	1.40	5.22
Sweets	−0.64	1.54	7.15
Meat	−3.57	1.57	1.17
Processed	4.06	1.77	2.53
Spices	2.02	1.82	2.12
Energy	13.15	3.95	–
Dairy	−2.06	7.28	14.70

Notes: Categories are sorted by reallocated expenditure. Consumption floor in per capita monthly expenditure is the estimated basic necessary quantity τ_i multiplied by average price for the category i. Reallocation in per capita monthly expenditure is the difference in expenditure spent on i when exposed to 1 SD more inequality (0.14) compared to the representative household (equation 8). Reallocation in per capita daily calories is the daily calorie equivalent of the expenditure reallocation using the average calorie content per Indian rupee (Online Appendix Table A.7).

are used abundantly by poorer populations in reaction to a sense of deprivation. Processed food and drinks are consumed at social events and festivals. Meat is an celebratory meal that is not taboo for a significant portion of poor households belonging to Hindu scheduled castes and non-Hindu religions.[48] For the representative household, most of these relative necessities have a small or insignificant consumption floor.

Poor households are ready to give up a substantial amount of calories to afford relative necessities. We find that an increase of 1 SD in inequality exposure is associated with 186 fewer calories per day per capita in total (sum of the third column of Table 4). This amounts to nearly one and a half adult portions of cooked rice (100g, 130 calories). The calorie Engel curve estimation was closer to 1 portion, but the 186 calories remain within the CI of the 2SLS parametric estimate. It could be argued that some of the relative necessities, such as dairy products, are conducive to a diversification of the diet of the poor. However, relative necessities appear to fit nutritional needs poorly in the context of dire malnutrition faced by households living below the poverty line. In terms of both calorie and protein content, it would be optimal to spend an additional Indian rupee on cereals instead. One Indian rupee of cereals has more than five times the calorie content of one Indian rupee of dairy products, but also more than four times the protein content (Online Appendix Table A.7).

48. Households might hesitate to disclose consumption of culturally or religiously sensitive items, such as meat or alcohol. This is not a primary issue in our context, as the majority of relative necessities do not fall into the category of taboo goods.

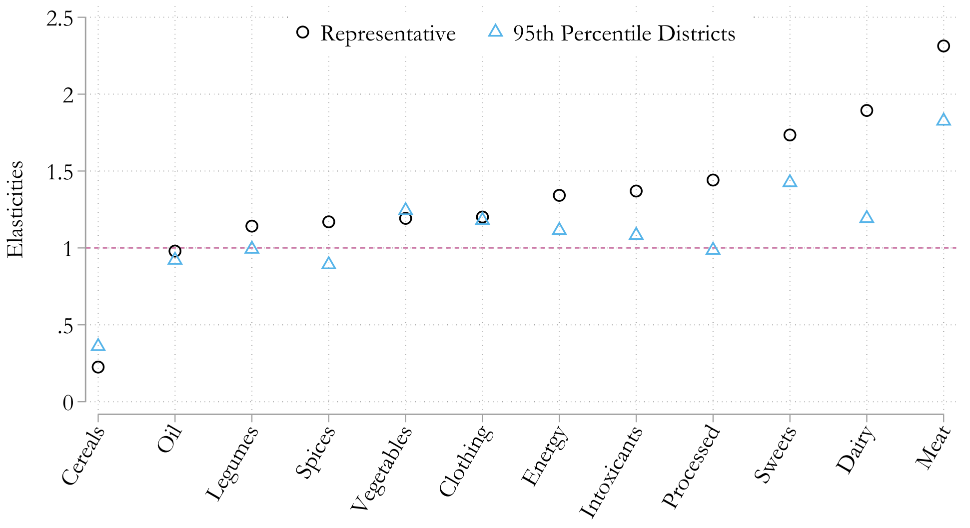

FIGURE 2. Change in income elasticities with inequality exposure: Income elasticities ξ_i by category estimated for a representative household and an equivalent household living in the 95th percentile of districts in terms of (average) relative deprivation.

How does inequality exposure affect income elasticities, or the hierarchy of needs? We calculate the elasticity for the representative household, and the elasticity for an equivalent household living in the 95th percentile of districts in terms of (average) relative deprivation, using equation (9) and the estimated parameters. Figure 2 illustrates the drastic difference in elasticities between these two households.

First, we confirm the hypothesis that relative necessities are luxuries (income elasticity above 1). The categories with the highest income elasticities for the representative household are all relative necessities responsible for much of the expenditure reallocation (meat, dairy products, sweets, processed food, and drinks). This pattern adds an interesting element to the finding of Heffetz (2011) that luxuries are goods perceived to be more conspicuous by the population, and therefore used by the rich to signal status. Our analysis shows that absolute luxuries are also relative necessities: The poor consider those goods to be more needed when they are exposed to richer people.

Second, a household living in the 95th percentile of districts most exposed to inequality faces a very different hierarchy of needs than the representative household. Elasticities of relative necessities shift downward, showing how these categories become relatively more necessary with inequality. Strikingly, two of them, spices and processed food and drinks, shift from a luxury to a normal good or a necessity (elasticity at or below 1). These substantial changes in elasticities underline the trade-off between social inclusion and adequate nutrition: relative necessities, which are both more expensive and less nutritious, amount to a greater share of poor households' budget when exposed to inequality.

TABLE 5. Calorie loss and malnutrition with inequality exposure.

	1993–1994	1999–2000	2004–2005	2009–2010
Daily per capita calories	1713.63	1643.00	1620.33	1610.80
Inequality exposure	0.42	0.41	0.45	0.47
Estimated calorie loss	553.75	544.19	593.90	624.26
% under malnutrition, actual	0.94	0.96	0.97	0.98
% under malnutrition, counterfactual	0.60	0.67	0.65	0.63

Notes: Malnutrition is calculated using the standard Indian thresholds: daily per capita consumption below 2,100 (urban) or 2,400 (rural) calories.

6.4. Inequality and the Caloric Consumption Puzzle

More than 90% of the households living below the poverty line in our data are malnourished, and this fraction increases over time. Deaton and Drèze (2009) were the first to observe what they called the "caloric consumption puzzle," which is that calorie consumption has declined in the past decades of growth for the entire Indian population. The main hypothesis for this decline is an improving disease environment, which lowers nutritional needs. However, in their empirical exploration of this hypothesis, Duh and Spears (2017) found that it accounts for only about one-fifth of the calorie decline. It is also arguably not the main cause of decline for poor and malnourished households.[49]

The strength and robustness of our findings suggest that relative deprivation may have been an overlooked yet significant alternative explanation for the caloric consumption puzzle, especially among the poor. Table 5 shows that the mean calorie consumption of BPL households in our data is below the malnutrition threshold and decreases over time. In contrast, their average inequality exposure increases over time, both according to our index (0.42–0.47) and other studies of inequality in India (Banerjee and Piketty 2005; Chancel and Piketty 2019). We produce counterfactual calorie intake for each household by adding to their actual consumption the calorie loss estimated from their inequality exposure. This counterfactual is equivalent to calculating calorie consumption in a situation where they would not be exposed to inequality. We find that the mean calorie loss with inequality exposure goes from 554 in 1993–1994 to 624 in 2009–2010. The direct effect of inequality on needs account for more than two–thirds of the mean calorie decline over time among poor households. The fraction of BPL households with malnutrition would be about 30 percentage points lower if they consumed these additional calories.

49. An alternative explanation is the shift over time to non-agricultural and less physically intensive occupations. We show in Table 1 that the addition of occupation-round fixed effects does not affect the calorie loss of the poor driven by inequality.

6.5. Robustness Specifications

We estimate three additional specifications to test the robustness of our results. First, to alleviate the concern that substitution patterns may affect our results, we estimate a generalization of the demand system including full cross-price substitution. The generalization, developed in Online Appendix Section E, loses in ease of interpretation as well as tractability given the high number of cross-price parameters. It nonetheless accommodates the same testable prediction. Second and third, we estimate the demand system on the entire sample of surveyed households (401,324 observations), and on the sample of non-poor households. These estimations may compromise certain assumptions of the LES demand system, such as the assumption of quasi-homotheticity of demand (i.e., Engel curves linear in total expenditure, see Online Appendix Figure B.1). They may also capture a different mechanism of inequality exposure on less deprived households. They are nonetheless a useful exercise of comparison to the baseline results.

The estimated basic necessary quantities are relatively similar across specifications. Online Appendix Figure F.3 compares the basic necessary quantities across the two robustness specifications of the LES demand system using the sample of all households and non-poor households (the generalized demand system does not lead to a straightforward estimation of basic necessary quantities as τ_i). It is striking that the estimates for basic necessary quantities are stable across samples, with an increase in categories with the highest basic quantities such as legumes, cooking oil, energy use, and cereals for non-poor (and, hence, all) households.

Relative necessities are consistent across robustness specifications too. Online Appendix Figure F.4 compares the reallocated expenditure by category with 1 SD increase in inequality exposure between the baseline specification, the generalized demand system with more flexibility in the price space, the sample of all households, and the sample of non-poor households. The results with the generalized demand system are both quantitatively and qualitatively similar. The same pattern also emerges when estimating demand on the sample of all households and non-poor households, with three notable differences: First, dairy products are not relative necessities for non-poor households, potentially implying that dairy products do not possess a distinguishing social feature after a certain level of income. Meat, however, does, and more so than for poor households. Third, intoxicants are the second highest amount of expenditure reallocation. For all specifications, inequality exposure makes nutritious categories such as cereals and vegetables relatively less necessary.

7. Alternative Channels

We documented a consistent and robust effect of inequality on the estimation of basic needs. This demand-side effect aligns with relative deprivation theory: It is driven by upward-looking comparisons and affects the hierarchy of needs, elevating the perceived necessity of luxuries at the expense of adequate nutrition. The estimated relative necessities corroborate with a relative deprivation channel: They are associated with social inclusion and shared in social settings in the Indian context. However, there may be alternative explanations consistent with the documented effect. Using internal and external data sources, we separately test three alternative channels of the effect of inequality on basic needs: access to credit markets, food security networks (Public Distribution System), and gift giving. We detail their potential effect in our setting, and confirm that these alternative effects are unlikely to explain our results.

7.1. Credit Markets

Inequality could causally affect poor households' credit access (Coibion et al. 2020), which, in turn, may affect intra-temporal consumption (Agarwal, Mikhed, and Scholnick 2020). We first test the link between inequality and credit between and within districts in our setting.[50] The NSS consumption surveys do not record household debt, but the NSS employment surveys report household outstanding debt and total monthly per capita expenditure for rural areas. Column 1 of Online Appendix Table G.1.1 reveals a positive correlation between how exposed to inequality poor rural households are on average between district, and district average credit. However, this correlation vanishes in column 2 after instrumenting for inequality exposure. Within districts, households more exposed to inequality also report higher credit, although the correlation is much weaker than between districts (column 3), and is not significant in the 2SLS estimation (column 4).[51] These results indicate that most of the link between inequality exposure and credit is correlational and possibly supply-driven.

Could such a link still affect intra-temporal consumption allocation? We match the average debt of BPL households in employment surveys to consumption surveys at the rural district-round level (the two sets of surveys cannot be matched at the household level). Online Appendix Table G.1.2 finds no significant correlation, and a near-zero point estimate, between credit and total per capita expenditure on our 12 categories (Online Appendix Table G.1.2, column 1). Replicating the analysis with total per capita expenditure on all other expenditures (durables, services, and toiletries), we find a positive correlation (column 2), driven by services (column 4), consistent with credit constraints on lumpy investments (e.g., health, education services). Decomposing day-to-day expenditure into necessities and luxuries using the same classification as in Section 4, we see a small negative effect on necessities (column 6), suggesting substitution toward the funding of services, but no significant debt effect on luxury consumption (column 7).

50. The within district analysis includes district fixed effects to absorb potential supply-side effects (e.g., an active banking system) affecting all households in the district.

51. Our instrument strategy addresses the same concerns raised in Section 4.2.2 about reverse causation (e.g., inequality driven by households over-borrowing for luxury consumption) and omitted variables (e.g., wealthier migrants or striving businesses drawn to regions with a more developed banking sector).

These tests indicate that the potential impact of inequality exposure on credit likely does not affect our results. If at all, households use debt for lumpy investments rather than small luxuries. Back-of-the-envelope calculations from Online Appendix Tables G.1.1 and G.1.2 suggest this credit channel may only explain a negligible fraction of the effect of inequality exposure on necessities.[52]

7.2. Food Security Networks

The Public Distribution System (PDS) is a network of "fair price" shops providing wheat, rice, sugar, and kerosene at subsidized rates in a limited monthly amount to poverty card holders.[53] Over time, concerns have been raised about leakages and corruption that render access to PDS shops difficult and divert parts of the stock to other populations than the ones intended (Overbeck 2016). Historical inequality may increase local corruption, hence, making PDS shops less accessible to the poor.

We explore this hypothesis by compiling expenditures on rice, wheat, sugar, and kerosene recorded from PDS source in all rounds. We test if the average amount distributed to BPL households in a district is correlated with district-level inequality exposure. We find no evidence of a negative effect of inequality on the distributed amount reported in our data (Online Appendix Table G.2.1). If anything, the results tend to point toward a positive correlation between inequality and access to PDS, although not precisely estimated and not significant once we instrument for total expenditure and inequality exposure. They confirm that BPL households do not have a more difficult access to subsidized items (and, in particular, cereals) in higher inequality areas.

7.3. Gift Giving

Inequality may also affect the level of charitable giving or pro-social behavior within a district (Andreoni, Nikiforakis, and Stoop 2017), for instance, through gift giving. In particular, wealthier households may give away items they consume themselves (luxuries) more than "cheap" goods. Online Appendix Table G.3.1 reports that charity and gift giving are rare occurrences of consumption.[54] The fraction of poor households in each round with recorded consumption coming from gifts or charity is too small to explain our main effects (below 0.3% of poor households for almost all categories).

52. Online Appendix Table G.1.1 shows a 1 SD increase in inequality exposure leads to a 28.5% debt increase ($P = 0.571$). Online Appendix Table G.1.2 indicates a 1% debt increase reduces necessity expenditure by 0.0058%, so a 1 SD increase in inequality exposure results in a 0.165% fall in necessity expenditure, $<$2% of the effect in Table 2.

53. All prior estimations include PDS expenditures in the consumption of each household and in the computation of local price indexes.

54. The consumption surveys record consumption from all sources, but the specific source "charity/gifts" is only reported in rounds 2004–2005 and 2009–2010 (for all categories except clothing). All prior estimations include charity/gifts consumption for each household.

Although it concerns a very small fraction of poor households, we test whether receiving gifts or charity is correlated with inequality. We run a linear probability model with the dependent variable being an indicator for receiving charity/gifts in each category (Online Appendix Table G.3.2). Controlling for log total per capita expenditure, inequality exposure seems positively associated, although not significantly for all categories, with receiving charity/gifts. There is no clear pattern that inequality exposure increases the probability to receive charity/gifts more for luxuries than for necessities. Interestingly, once we instrument for inequality exposure and total expenditure (Online Appendix Table G.3.3), the effect of inequality exposure for all categories becomes largely insignificant.

8. Conclusion

Using detailed information on the consumption choices of a large sample of poor households, we show that inequality affects the needs of the poor. Building on an Engel curves framework, we first document that relatively deprived households increase their consumption of luxury goods at the expense of calorie intake. Our analysis reveals that the observed calorie loss stems from comparisons with wealthier individuals, consistent with relative deprivation theory. We then estimate a model of demand where inequality exposure affects the needs of the poor. We provide evidence that inequality exposure drives the poor to consider luxuries as more necessary. In the Indian context, these "relative necessities" are expensive calories or non-caloric goods. The need-based channel is sizable: We estimate that the shift in the hierarchy of needs results in a major calorie loss, accounting for two-thirds of the calorie consumption puzzle in India for poor households.

We focused on isolating and quantifying a need-based channel, separately from supply-side factors. The consistency of calorie loss calculations found in both Engel curves and structural analysis highlights that the impact of inequality on the consumption floor is mainly demand-driven. While identifying the precise psychological mechanisms behind the shift in consumer needs among the less affluent falls beyond the scope of this paper, it provides avenues for future research. Two mechanisms, in particular, could be considered: First, relatively deprived households may purchase luxuries as a means to elevate their social standing (Heffetz 2011; Bursztyn et al. 2017). Second, an important strand of psychology literature points to negative psychological consequences of inequality exposure ("status anxiety"; Pickett and Wilkinson 2015; Case and Deaton 2021). The preference for socially visible goods that signify higher status, such as dairy products and energy, over goods that provide immediate hedonic value like sweets and intoxicants, aligns more closely with the first mechanism, the desire of relatively deprived individuals to regain a sense of community and inclusion.

Recent events with major distributional consequences, from the Great Recession to the COVID-19 pandemic, led to intense debates regarding the definition and regulations of so-called "essential" and "non-essential" products. Our findings suggest that when the poor are exposed to high levels of inequality, making luxury goods more expensive (e.g., via taxes) may have detrimental consequences on their standard of living. Instead, policies focused on reducing inequality seem more promising avenues. More generally, our results provide a rationale for the (apparently) wasteful behavior of the poor, and call for the systematic inclusion of relative needs in the design of poverty alleviation policies.

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Supplementary Material

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